

Chapter 8 Practice Test Noncalculator #1-10 (25min) Calculator #11-19 (30min)

1. Evaluate: $\int_2^{\infty} \frac{2dx}{(x-1)^2}$

- a) -2 b) $-\frac{1}{2}$ c) $\frac{1}{2}$ d) 2 e) ∞

2. $\int x^4 \ln x dx =$

- a) $x^4 - \frac{4}{3}x^3 + C$ b) $\frac{1}{25}x^5 [5\ln x - 1] + C$ c) $x^3 + \ln 4x^4 + C$ d) $x^3 [1 + 4\ln x] + C$ e) $x^3 - \ln 5x^5 + C$

3. $\int_4^5 \frac{dx}{x^2 - 5x + 6} =$

- a) $\ln \left| \frac{1}{6} \right|$ b) $\ln \left| \frac{1}{3} \right|$ c) $\ln \left| \frac{3}{4} \right|$ d) $\ln \left| \frac{4}{3} \right|$ e) $\ln |3|$

4. Evaluate the following for any positive constant k : $\int_{-\infty}^{\infty} \frac{dx}{x^2 + k^2}$

- a) $-\frac{\pi}{k}$ b) 0 c) $\frac{\pi}{k}$ d) $\frac{\pi}{2k}$ e) divergent

5. Evaluate the following given c is a constant: $\lim_{x \rightarrow 0} \frac{c \tan cx}{x^2 + 8x}$

- a) 0 b) $\frac{1}{8}$ c) $\frac{c^2}{8}$ d) c^2 e) DNE

6. $\int_3^7 \frac{x+1}{x^2 + 2x - 3} dx =$

- a) $\frac{1}{2} \ln 2$ b) $\frac{1}{2} \ln 5$ c) $\frac{3}{2} \ln 2$ d) $\ln 5$ e) $2 \ln 5$

7. The derivative of a function f is given by $f'(x) = \ln x$ for $x > 0$ and $f(1) = -2$. Find $f(e)$.

- a) $-e - 2$ b) -2 c) -1 d) $e - 2$ e) e

8. Evaluate: $\int x \sec x \tan x dx$

a) $\frac{x^2}{2} \sec x + C$

b) $x \sec x - \sec x \tan x + C$

c) $x \sec x + \ln|\sec x + \tan x| + C$

d) $x \sec x - \ln|\sec x + \tan x| + C$

e) $x \tan x + \ln|\cos x| + C$

9. $\int_0^{\sqrt{3}} \frac{x}{\sqrt{4-x^2}} dx$

a) -1

b) $\frac{\pi}{6}$

c) 1

d) $\frac{\pi}{3}$

e) 2

10. $\int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$

a) -1

b) $\frac{\pi}{6}$

c) 1

d) $\frac{\pi}{3}$

e) 2

11. Evaluate: $\int \frac{11x+17}{2x^2+7x-4} dx$

a) $\frac{5}{2} \ln|2x-1| + 3 \ln|x+4| + C$

b) $5 \ln|2x-1| + 3 \ln|x+4| + C$

c) $10 \ln|2x-1| + 3 \ln|x+4| + C$

d) $5 \ln|2x-4| + 3 \ln|x-1| + C$

e) $3 \ln|2x-1| + 5 \ln|x+4| + C$

12. $\int_{-3}^0 \frac{dx}{\sqrt[3]{x+3}}$

a) $\frac{2}{3}(3)^{\frac{3}{2}}$

b) $(3)^{\frac{3}{2}}$

c) $\frac{3}{2}(3)^{\frac{3}{2}}$

d) $\frac{3}{2}(3)^{\frac{2}{3}}$

e) divergent

13. $\int \frac{x^2}{x^2-4} dx =$

a) $x + \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$

b) $x - \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$

c) $x + \ln\left|\frac{x-2}{x+2}\right| + C$

d) $x + \ln\left|\frac{x+2}{x-2}\right| + C$

e) $x - \ln\left|\frac{x-2}{x+2}\right| + C$

14. Evaluate: $\int x e^{2x} dx$

- a) $2e^{2x}(x-2)+C$ b) $x^2 e^{x^2} + C$ c) $\frac{e^{2x}}{4}(2x-1)+C$ d) $\frac{x^2 e^{x^2}}{2} + C$ e) $e^{2x}(x-1)+C$

15. $\int_1^{\infty} \frac{1}{x(\ln x + 1)^2} dx =$

- a) 0.5 b) 1 c) 2 d) 3 e) ∞

16. $\int_1^9 \frac{dx}{\sqrt[3]{(x-1)^5}} =$

- a) -6 b) $-\frac{3}{8}$ c) 0 d) $\frac{5}{8}(8)^{\frac{2}{5}}$ e) DNE

17. Find: $\lim_{x \rightarrow 0} \frac{\int_0^x (e^{t^2} - 1) dt}{2x^2}$

- a) $-\frac{1}{2}$ b) 0 c) $\frac{1}{2}$ d) $\frac{e}{2}$ e) DNE

18. Let f be the function defined by $f(x) = \frac{1}{x^2 - x}$ for $x \geq 2$ and let R be the region between the graph of f and the x -axis. Determine the area of region R.

- a) 1 b) $\ln 2$ c) 2 d) $2 \ln 2$ e) Divergent

19. Solve the following differential equation for $y = f(x)$ given $\frac{dy}{dx} = xy \cos x$ and $f(\pi) = 1$.

- a) $e^{x \sin x + \cos x}$ b) $e^{x \sin x + \cos x + 1}$ c) $e^{x \sin x - \cos x + 1}$ d) $e^{x \cos x + \sin x + \pi}$ e) $\sqrt{2x \sin x + 2 \cos x + 3}$

Free Response 1 - NC

Consider the region under the curve $f(x) = xe^{-\frac{x}{2}}$, above the x -axis, on the interval $[0, \infty)$.

- Over what interval(s) is f increasing? Justify your answer.
- Find the absolute extrema of f over the interval $[0, \infty)$.
- Write, but do not evaluate, an integral expression to find the volume of the solid obtained by revolving the region about the x -axis.
- Find the area of the region.

Free Response 2 - NC

BC2: The piecewise defined function h , where k is a constant is given by

$$h(x) = \begin{cases} 3h(x^2 - 6), & x \leq -2 \\ \frac{kx - 1}{(x - 1)(x + 2)}, & -2 < x < 2 \\ \frac{12}{x^2}, & x \geq 2 \end{cases}$$

- Find the value of k that makes $h(x)$ continuous at $x = 2$.
- Evaluate $\lim_{x \rightarrow -3} \frac{h(x) - 4}{x^2 - 9}$ or state that it does not exist. Show the work that leads to your answer.
- Evaluate $\int_2^{\infty} h(x) dx$ or show that it diverges. Show the work that leads to your answer.
- Let $k = 4$. Evaluate $\int_{-1}^0 h(x) dx$. Show the work that leads to your answer.

Answers

1. With an infinite bound, $\lim_{b \rightarrow \infty} \int_2^b \frac{2dx}{(x-1)^2}$. This requires a u-sub with $u = x-1, du = dx$ so $\lim_{b \rightarrow \infty} 2 \int_2^b \frac{du}{u^2}$. So,
 $\lim_{b \rightarrow \infty} \left[-\frac{2}{x-1} \right]_2^b = \lim_{b \rightarrow \infty} \left[-\frac{2}{b-1} + \frac{2}{1} \right]$. This first limit goes to 0, so the integral goes to 2, so **D**.
2. This is integration by parts and ILATE says $u = \ln x$, $dv = x^4 dx$. So $du = \frac{1}{x} dx$, $v = \frac{1}{5} x^5$ and
 $\int x^4 \ln x dx = \frac{1}{5} x^5 \ln x - \int \frac{1}{5} x^5 \frac{1}{x} dx$. Simplifying the integrand gives us $\frac{1}{5} x^5 \ln x - \frac{1}{5} \int x^4 dx$ which can be integrated as is, so $\frac{1}{5} x^5 \ln x - \frac{1}{25} x^5 + C$, which can be rewritten as **B**.
3. Since the denominator can be factored, we will use partial fractions. So $\frac{1}{x^2 - 5x + 6} = \frac{A}{x-2} + \frac{B}{x-3}$ or
 $1 = A(x-3) + B(x-2)$. Using $x = 3$, $x = 2$ gives us $B = 1$, $A = -1$, so
 $\int_4^5 \frac{1}{x^2 - 5x + 6} dx = \int_4^5 \frac{-1}{x-2} dx + \int_4^5 \frac{1}{x-3} dx$. These can be integrated as with 2 u-substitutions that don't require
any constants, so $-\ln|x-2| + \ln|x-3| \Big|_4^5 = \ln \left| \frac{x-3}{x-2} \right| \Big|_4^5 = \ln \left| \frac{2}{3} \right| - \ln \left| \frac{1}{2} \right| = \ln \left| \frac{4}{3} \right|$ so **D**.
4. Since both bounds are infinite, we must pick a constant, so 0. So, $\int_{-\infty}^{\infty} \frac{dx}{x^2 + k^2} = \lim_{b \rightarrow -\infty} \int_b^0 \frac{dx}{x^2 + k^2} + \lim_{c \rightarrow \infty} \int_0^c \frac{dx}{x^2 + k^2}$
where both are arctan. In both of these, $u = x, du = dx$, $a = k$, and $\lim_{b \rightarrow -\infty} \frac{1}{k} \arctan \left(\frac{x}{k} \right) \Big|_b^0 + \lim_{c \rightarrow \infty} \frac{1}{k} \arctan \left(\frac{x}{k} \right) \Big|_0^c$.
Evaluating gives us $\lim_{b \rightarrow -\infty} \left(\frac{1}{k} \arctan(0) - \frac{1}{k} \arctan \left(\frac{b}{k} \right) \right) + \lim_{c \rightarrow \infty} \left(\frac{1}{k} \arctan \left(\frac{c}{k} \right) - \frac{1}{k} \arctan(0) \right)$. Therefore,
 $\lim_{b \rightarrow -\infty} \left(-\frac{1}{k} \arctan \left(\frac{b}{k} \right) \right) + \lim_{c \rightarrow \infty} \left(\frac{1}{k} \arctan \left(\frac{c}{k} \right) \right)$ and $\left(-\frac{1}{k} \left(-\frac{\pi}{2} \right) \right) + \left(\frac{1}{k} \left(\frac{\pi}{2} \right) \right) = \frac{\pi}{2k} + \frac{\pi}{2k} = \frac{\pi}{k}$ so **C**.
5. Before using L'Hôpital's Rule, check that the limit is indeterminate $\frac{0}{0}$, which it is. So,
 $\lim_{x \rightarrow 0} \frac{c \tan cx}{x^2 + 8x} = \lim_{x \rightarrow 0} \frac{c \sec^2 cx \bullet c}{2x + 8} = \lim_{x \rightarrow 0} \frac{c^2 \sec^2 cx}{2x + 8}$. Re-evaluating this gives us $\frac{c^2 \bullet \sec^2 0}{8} = \frac{c^2}{8}$ or **C**.

6. Since the denominator can be factored, we will use partial fractions. So $\frac{x+1}{x^2+2x-3} = \frac{A}{x-1} + \frac{B}{x+3}$ or $x+1 = A(x+3) + B(x-1)$. Using $x = -3$, $x = 1$ gives us $-2 = -4B$, $2 = 4A$, so $\frac{1}{2} = B$, $\frac{1}{2} = A$ and $\int_3^7 \frac{x+1}{x^2+2x-3} dx = \frac{1}{2} \int_3^7 \frac{1}{x-1} dx + \frac{1}{2} \int_3^7 \frac{1}{x+3} dx$. These can be integrated as with 2 u-substitutions that don't require any constants, so $\left. \frac{1}{2} \ln|x-1| + \frac{1}{2} \ln|x+3| \right|_3^7 = \frac{1}{2} \ln|(x-1)(x+3)| \Big|_3^7 = \frac{1}{2} \ln|60| - \frac{1}{2} \ln|12| = \frac{1}{2} \ln \left| \frac{60}{12} \right|$ so **B**.
7. Given f' , to find f , we must integrate. We can either use a definite integral and add on -2 or use an indefinite integral, use the initial condition, then find $f(e)$. Using the first, $f(e) = -2 + \int_1^e \ln x dx$ and this is integration by parts and ILATE says $u = \ln x$, $dv = dx$. So $du = \frac{1}{x} dx$, $v = x$ and $-2 + \left[x \ln x - \int_1^e \left(\frac{1}{x} \right) x dx \right]$. Simplifying the integrand gives us $-2 + \left[x \ln x - \int_1^e dx \right]$ which can be integrated as is, so $-2 + [x \ln x - x]_1^e = -2 + [(e \ln e - e) - (\ln 1 - 1)]$, or $-2 + [(e - e) - (-1)] = -2 + [0 + 1] = -1$ so **C**.
8. This is integration by parts and ILATE says $u = x$, $dv = \sec x \tan x dx$. So $du = dx$, $v = \sec x$ and $\int x \sec x \tan x dx = x \sec x - \int \sec x dx$. This can be integrated as is, so $x \sec x - \ln|\sec x + \tan x| + C$, so **D**.
9. Although the denominator looks like arcsin, it does not work. This is a u-substitution where $u = 4 - x^2$, $du = -2x dx$. So $-\frac{1}{2} \int_0^{\sqrt{3}} \frac{du}{\sqrt{u}} = -\frac{1}{2} \left[2\sqrt{4-x^2} \Big|_0^{\sqrt{3}} \right] = -\sqrt{4-x^2} \Big|_0^{\sqrt{3}}$ or $-\sqrt{1} - (-\sqrt{4}) = -1 + 2 = 1$ or **C**.
10. This one is arcsin where $u = x$, $du = dx$, $a = 2$. So $\int_0^{\sqrt{3}} \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\left(\frac{x}{2}\right) \Big|_0^{\sqrt{3}} = \arcsin\left(\frac{\sqrt{3}}{2}\right) - \arcsin(0)$. Evaluating this gives us $\frac{\pi}{3}$ or **D**.
11. Since the denominator can be factored, we will use partial fractions. So $\frac{11x+17}{2x^2+7x-4} = \frac{A}{2x-1} + \frac{B}{x+4}$ or $11x+17 = A(x+4) + B(2x-1)$. Using $x = -4$, $x = \frac{1}{2}$ gives us $-27 = -9B$, $\frac{45}{2} = \frac{9}{2}A$, so $3 = B$, $5 = A$ and $\int \frac{11x+17}{2x^2+7x-4} dx = 5 \int \frac{1}{2x-1} dx + 3 \int \frac{1}{x+4} dx$. These can be integrated with 2 u-substitutions where only the first needs a constant, so $u = 2x-1$, $du = 2dx$. So, $\frac{5}{2} \int \frac{2}{2x-1} dx + 3 \int \frac{1}{x+4} dx$ and $\frac{5}{2} \ln|2x-1| + 3 \ln|x+4| + C$ so **A**.

12. Since the lower bound is a domain restriction, $\lim_{b \rightarrow -3} \int_b^0 \frac{dx}{\sqrt[3]{x+3}}$. This requires a basic u-substitution that

doesn't need a constant. So, $\lim_{b \rightarrow -3} \int_b^0 u^{-\frac{1}{3}} du = \lim_{b \rightarrow -3} \frac{3}{2} (x+3)^{\frac{2}{3}} \Big|_b^0 = \lim_{b \rightarrow -3} \left[\frac{3}{2} (3)^{\frac{2}{3}} - \frac{3}{2} (b+3)^{\frac{2}{3}} \right]$. As $b \rightarrow -3$,

$\frac{3}{2} (b+3)^{\frac{2}{3}}$ will go to 0, so $\lim_{b \rightarrow -3} \left[\frac{3}{2} (3)^{\frac{2}{3}} - \frac{3}{2} (b+3)^{\frac{2}{3}} \right] = \frac{3}{2} (3)^{\frac{2}{3}}$ or **D**.

13. When the powers are equivalent, do long division. So, $\int \frac{x^2}{x^2-4} dx = \int \left(1 + \frac{4}{x^2-4} \right) dx$, which we should

separate to be $\int 1 dx + \int \frac{4}{x^2-4} dx$. The second integral can be factored, so we will try partial fractions

with $\frac{4}{x^2-4} = \frac{A}{x-2} + \frac{B}{x+2}$ or $4 = A(x+2) + B(x-2)$. Using $x = -2$, $x = 2$ gives us $4 = -4B$, $4 = 4A$, so

$-1 = B$, $1 = A$ and $\int 1 dx + 1 \int \frac{1}{x-2} dx - 1 \int \frac{1}{x+2} dx$. These can be integrated with 2 u-sub where neither

requires a constant. So, $x + \ln|x-2| - \ln|x+2| + C = x + \ln \left| \frac{x-2}{x+2} \right| + C$ so **C**.

14. This is integration by parts and ILATE says $u = x$, $dv = e^{2x} dx$. The integration part of v from dv

requires a u-substitution ($u = 2x$, $du = 2dx$), so $du = dx$, $v = \frac{1}{2} e^{2x}$ and $\int x e^{2x} dx = \frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx$.

This integral requires the same u-substitution $u = 2x$, $du = 2dx$. So $\frac{1}{2} x e^{2x} - \frac{1}{2} \left(\frac{1}{2} e^{2x} \right) + C$, or

$\frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$, which can be rewritten as $\frac{1}{4} e^{2x} (2x-1) + C$ or **C**.

15. With an infinite bound, $\lim_{b \rightarrow \infty} \int_1^b \frac{1 dx}{x(\ln x + 1)^2}$. This requires a u-sub with $u = \ln x + 1$, $du = \frac{1}{x} dx$ so $\lim_{b \rightarrow \infty} \int_1^b \frac{du}{u^2}$.

So, $\lim_{b \rightarrow \infty} \left[-\frac{1}{\ln x + 1} \right]_1^b = \lim_{b \rightarrow \infty} \left[-\frac{1}{\ln b + 1} + \frac{1}{\ln 1 + 1} \right]$. This first limit goes to 0, and the integral goes to 1, so **B**.

16. Since the lower bound is a domain restriction, $\lim_{b \rightarrow 1} \int_b^9 (x-1)^{-\frac{5}{3}} dx$. This requires a basic u-substitution that

doesn't need a constant. So, $\lim_{b \rightarrow 1} \int_b^9 u^{-\frac{5}{3}} du = \lim_{b \rightarrow 1} -\frac{3}{2} (x-1)^{-\frac{2}{3}} \Big|_b^9 = \lim_{b \rightarrow 1} \left[-\frac{3}{2(8)^{\frac{2}{3}}} + \frac{3}{2(b-1)^{\frac{2}{3}}} \right]$. As $b \rightarrow 1$,

$2(b-1)^{\frac{2}{3}}$ will go to 0, which is in the denominator. Since $\frac{3}{0}$ is undefined, the limit does not exist so **E**.

17. Before using L'Hôpital's Rule, check that the limit is indeterminate, which it is. So, taking the derivative

of the numerator is using the 2nd FTC, which gives us $\lim_{x \rightarrow 0} \frac{\int_0^x (e^{t^2} - 1) dt}{2x^2} = \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1) \cdot 1 - (e^{0^2} - 1) \cdot 0}{4x}$. This

simplifies to be $\lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{4x}$ and re-evaluating this gives us indeterminate again. So, another round of

L'Hôpital's Rule gives us $\lim_{x \rightarrow 0} \frac{e^{x^2}(2x)}{4}$. Re-evaluating this gives us $\frac{0}{4} = 0$ (not indeterminate) or **B**.

18. To find area, we must integrate the top curve – the bottom curve. For all $x \geq 2$, f is on top. So the area

is $\int_2^\infty \frac{1}{x^2 - x} dx$ and with an infinite bound, we write $\lim_{b \rightarrow \infty} \int_2^b \frac{dx}{x^2 - x}$. Since the denominator can be factored,

we will use partial fractions. So $\frac{1}{x^2 - x} = \frac{A}{x} + \frac{B}{x - 1}$ or $1 = A(x - 1) + Bx$. Using $x = 1$, $x = 0$ gives us

$1 = 1B$, $1 = -1A$, so $1 = B$, $-1 = A$ and $\lim_{b \rightarrow \infty} \int_2^b \frac{dx}{x^2 - x} = \lim_{b \rightarrow \infty} \left[\int_2^b \frac{dx}{x - 1} - \int_2^b \frac{dx}{x} \right]$. These can be integrated with 2 u-

subs where neither need a constant. So, $\lim_{b \rightarrow \infty} [\ln|x - 1| - \ln|x|]_2^b = \lim_{b \rightarrow \infty} \left[\ln \left| \frac{x - 1}{x} \right| \right]_2^b$, which can be rewritten as

$\lim_{b \rightarrow \infty} \left[\ln \left| 1 - \frac{1}{x} \right| \right]_2^b$. Evaluating the bounds gives us $\lim_{b \rightarrow \infty} \left[\ln \left| 1 - \frac{1}{b} \right| - \ln \left| \frac{1}{2} \right| \right]$ with the limit going to $\ln 1 - \ln \left| \frac{1}{2} \right|$.

This goes to $0 - \ln \left| \frac{1}{2} \right| = -\ln \left| \frac{1}{2} \right| = \ln \left(\frac{1}{2} \right)^{-1} = \ln 2$ so **B**.

19. To solve for $y = f(x)$ given the derivative, we must integrate by separating the variables. So,

$\int \frac{dy}{y} = \int x \cos x dx$ with the second integral requiring an integration by parts. ILATE says that

$u = x$, $dv = \cos x dx$ and $du = dx$, $v = \sin x$ meaning $\int \frac{dy}{y} = x \sin x - \int \sin x dx$. Both of these integrals

can be integrated as is, so $\ln|y| = x \sin x + \cos x + C$. Using $f(\pi) = 1$ to find C gives us

$\ln|1| = \pi \sin \pi + \cos \pi + C$ or $0 = 0 - 1 + C$ and $C = 1$. So, $\ln|y| = x \sin x + \cos x + 1$ and solving for y

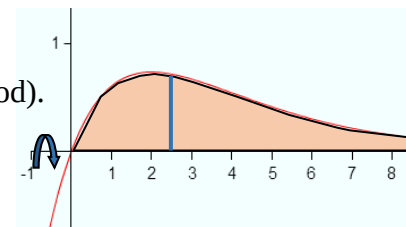
gives us $|y| = e^{x \sin x + \cos x + 1}$ and since the given y - value is positive, $y = e^{x \sin x + \cos x + 1}$ or **B**.

FR 1 a) f is increasing when $f' > 0$. So $f' = e^{-\frac{x}{2}} + xe^{-\frac{x}{2}}\left(-\frac{1}{2}\right) = e^{-\frac{x}{2}}\left(1 - \frac{1}{2}x\right)$ and the critical values are when $f' = 0$, und. This occurs at $x = 2$ and the line test shows that f is increasing from $(0, 2)$ since $x > 0$.

b) Absolute extrema occur at critical values ($x = 2$) and endpoints $[0, \infty)$. So we need to evaluate f to determine the extrema, where $f(2) = 2e^{-1}$, $f(0) = 0$. We need a limit to evaluate $f(\infty)$, so $\lim_{x \rightarrow \infty} xe^{-\frac{x}{2}}$. Taking the limit gives us $\infty(0)$, which is indeterminate. So we can use L'Hôpital's Rule if we can get a fraction. So, $\lim_{x \rightarrow \infty} \left[\frac{x}{e^{\frac{x}{2}}} \right]$ and L'Hôpital's Rule gives us $\lim_{x \rightarrow \infty} \left[\frac{x}{e^{\frac{x}{2}}} \right] = \lim_{x \rightarrow \infty} \left[\frac{1}{0.5e^{\frac{x}{2}}} \right]$ which gives us 0. Therefore, the absolute max is $2e^{-1}$ and the absolute min is 0.

c) Based on info from a) and b), we can make a sketch. Since the region is completely against the axis of revolution, there is no inner radius (disc method). Since the region is revolved about the x-axis, everything is in terms of x .

The radius is curve – axis, so $f - 0$ and the volume is $V = \pi \int_0^{\infty} [f(x)]^2 dx$.



d) The description says that f is on top, so the area is $\int_0^{\infty} xe^{-\frac{x}{2}} dx$ and since one bound is infinite,

$\lim_{b \rightarrow \infty} \int_0^b xe^{-\frac{x}{2}} dx$. This is integration by parts and ILATE says $u = x$, $dv = e^{-\frac{x}{2}} dx$. The integration part of

v from dv requires a u-substitution $\left(u = -\frac{x}{2}, du = -\frac{1}{2} dx\right)$, so $du = dx$, $v = -2e^{-\frac{x}{2}}$ and

$\lim_{b \rightarrow \infty} \int_0^b xe^{-\frac{x}{2}} dx = \lim_{b \rightarrow \infty} \left[-2xe^{-\frac{x}{2}} + \int_0^b 2e^{-\frac{x}{2}} dx \right]$. This integral requires the same u-substitution

$\left(u = -\frac{x}{2}, du = -\frac{1}{2} dx\right)$. So $\lim_{b \rightarrow \infty} \left[-2xe^{-\frac{x}{2}} - 4e^{-\frac{x}{2}} \right]_0^b = \lim_{b \rightarrow \infty} \left[e^{-\frac{x}{2}}(-2x - 4) \right]_0^b$ and evaluating the bounds

give us $\lim_{b \rightarrow \infty} \left[e^{-\frac{b}{2}}(-2b - 4) \right] - (-4)$. Taking the limit gives us $0(-\infty)$, which is indeterminate. So we

can use L'Hôpital's Rule if we can get a fraction. So, $\lim_{b \rightarrow \infty} \left[e^{-\frac{b}{2}}(-2b - 4) \right] + 4 = \lim_{b \rightarrow \infty} \left[\frac{-2b - 4}{e^{\frac{b}{2}}} \right] + 4$ and

L'Hôpital's Rule gives us $\lim_{b \rightarrow \infty} \left[\frac{-2b - 4}{e^{\frac{b}{2}}} \right] + 4 = \lim_{b \rightarrow \infty} \left[\frac{-2}{0.5e^{\frac{b}{2}}} \right] + 4$ and the limit goes to 0, so the area is 4.

FR 2 a) Continuous means $\lim_{x \rightarrow 2} h(x) = h(2)$. Since h breaks at $x = 2$, we will use one-sided limits. So,

$$\lim_{x \rightarrow 2^-} h(x) = \frac{2k-1}{(1)(4)} = \frac{2k-1}{4}, \lim_{x \rightarrow 2^+} h(x) = \frac{12}{4} = 3 \text{ and } h(2) = \frac{12}{4} = 3. \text{ So to be continuous, } \frac{2k-1}{4} = 3, 2k-1 = 12, k = \frac{13}{2}.$$

b) $\lim_{x \rightarrow -3} \frac{h(x)-4}{x^2-9}$ and after direct substitution, $\lim_{x \rightarrow -3} h(x) - 4 = 3h(3) - 4 = 3\left(\frac{12}{9}\right) - 4 = 0$ and $\lim_{x \rightarrow -3} x^2 - 9 = 0$. This means that L'Hôpital applies. Therefore, $\lim_{x \rightarrow -3} \frac{h(x)-4}{x^2-9} = \lim_{x \rightarrow -3} \frac{h'(x)}{2x}$. As x approaches -3 , $h'(x) = 3h'(x^2-6) \cdot 2x$. So, $h'(x) = 6xh'(x^2-6)$ and $\lim_{x \rightarrow -3} h'(x) = -18h'(3)$. When x approaches 3 , $h'(x) = \frac{-24}{x^3}$. So $-18h'(3) = -18\left(\frac{-24}{27}\right) = 16$. So, $\lim_{x \rightarrow -3} \frac{h'(x)}{2x} = \frac{16}{-6} = -\frac{8}{3}$.

c) $\int_2^\infty h(x)dx = \lim_{b \rightarrow \infty} \int_2^b h(x)dx$. For all $x \geq 2$, $h(x)$ is only $\frac{12}{x^2}$. So, $\lim_{b \rightarrow \infty} \int_2^b \frac{12}{x^2} dx$ which can be rewritten as $\lim_{b \rightarrow \infty} \int_2^b 12x^{-2} dx$, which is a power rule. $\lim_{b \rightarrow \infty} \left. \frac{-12}{x} \right|_2^b = \lim_{b \rightarrow \infty} \left(\frac{-12}{b} - \frac{-12}{2} \right)$. This limit goes to 0, so $(0 + 6) = 6$.

d) From $[-1, 0]$, $h(x)$ is $\frac{kx-1}{(x-1)(x+2)}$. So, $\int_{-1}^0 \frac{4x-1}{(x-1)(x+2)} dx$ which means we can use partial fractions. So, $\frac{A}{x-1} + \frac{B}{x+2}$ and $A(x+2) + B(x-1) = 4x-1$ and using $x = 1$, $x = -2$ we get $A = 1$, $B = 3$ giving us $\int_{-1}^0 \frac{4x-1}{(x-1)(x+2)} dx = \int_{-1}^0 \left(\frac{1}{x-1} + \frac{3}{x+2} \right) dx$. After using u-sub, each $du = 1dx$ so the antiderivative is $\ln|x-1| + 3\ln|x+2| \Big|_{-1}^0$, so $(\ln|-1| + 3\ln|2|) - (\ln|-2| + 3\ln|1|) = 3\ln 2 - \ln 2 = 2\ln 2$.